

Assignment 1

DSUC+DBMS, 1st MCA, HBTU

Important Instructions:

1. Students can solve this assignment either individually or in groups. [A group of up to 5 members is recommended].
2. The size of the group cannot exceed five members.
3. Rules for applying AI tools:
 - a. AI tools are permitted.
 - b. Each individual student must clearly mention (i) the name of the AI tool used, (ii) the prompts given to AI tool, (iii) the output generated from each prompts after using AI tool and (iv) how the generated result/output is implemented in the code developed by the student. [detailed explanation is expected]
 - c. The student written code and AI generated code must be highlighted separately. [use comments or color coding to do this separation].
4. Rules for using search engines:
 - a. Students can take help from search engines like google.
 - b. All references must clearly be mentioned.
5. Rules for expert advice (except me):
 - a. Students can seek help from experts/seniors/friends.
 - b. Students must document the queries raised and the corresponding responses provided by experts/seniors/friends.
6. Assignment submission rules:
 - a. Deadline: 22/April/2026 (till 5:00 PM).
 - b. Late submission: 25/April/2026 (till 3:00 PM).
 - c. All late submissions are evaluated out of 15 marks.
 - d. No submission is allowed after 25/April/2026 (3:00 PM).
7. Assignment viva and live demo:
 - a. All students (submitting assignments on 22/April/2026) must present a live demonstration of their assignment work on the same day, either to me or to the team allocated for this purpose. The demonstrations will be conducted post-lunch. The exact schedule will be circulated shortly.
 - b. Same guidelines are applicable for late submissions on 25/April/2026.
8. Students will be marked based on correctness, innovation, beauty, implementation style, coding skills and other such parameters.
9. Use of AI tools is permitted but not recommended for skill enhancement.
10. Make use of graphs, figures, tables and other aids to make your answers and presentations attractive.
11. Only C programming language can be used coding. Codes written in any other programming languages will not be accepted, and using other languages will result in a zero for the assignment.

Data Structures (DS) provide ways to organize data in the main memory of the computer. On the other hand Databases (DB) help to organize data in secondary memory of the computer. Knowledge of DS and DB help to master fundamental computer science courses. This assignment helps learners to feel the real power of DS and DB.

Assignment Prerequisites

1. Install Oracle VirtualBox which is virtual machine or VM using link: <https://www.virtualbox.org/wiki/Downloads>
2. Install Alpine Linux (standard, x86_64 version, see figure 1) on Oracle VirtualBox as Alpine Linux Virtual Machine (or Alpine VM in short). Use alpine commands to setup alpine linux on your VM (see Note 1). The link for Alpine Linux: <https://alpinelinux.org/downloads/>

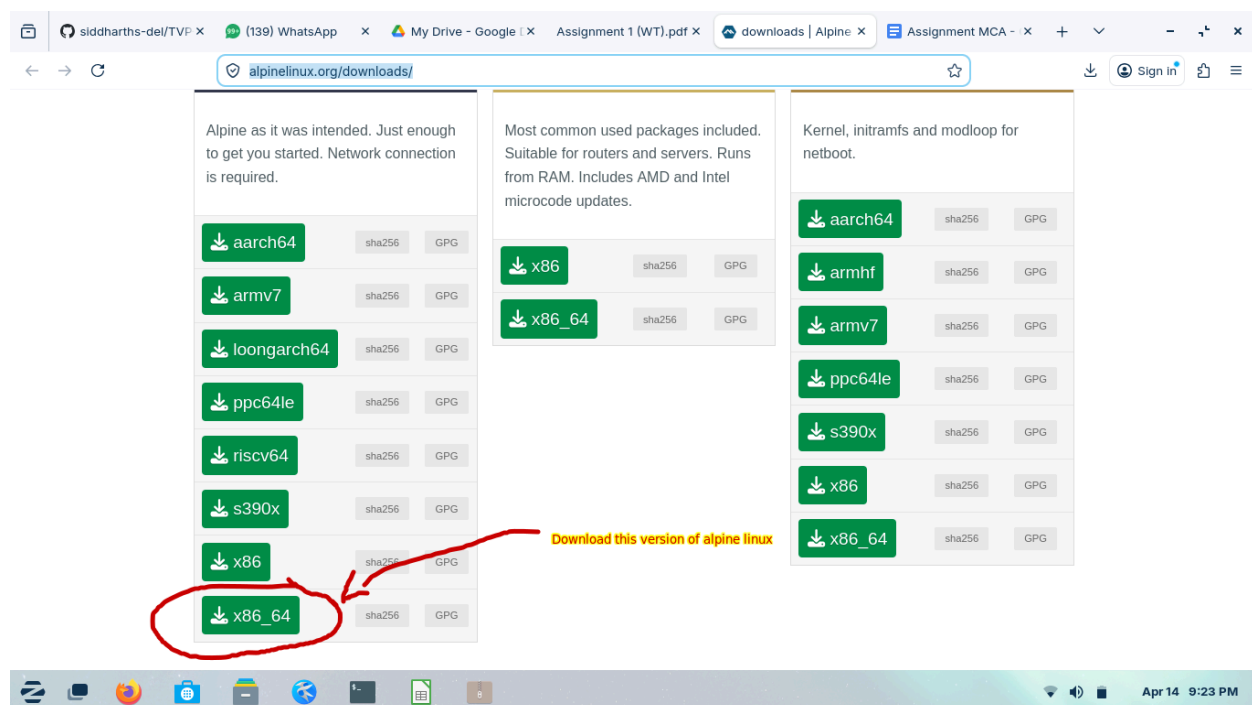


Figure 1: Version of Alpine Linux Chosen for Assignment

Note-1: Alpine Linux Operating System Handbook link for understanding OS commands and installation process is as follows: <https://docs.alpinelinux.org/user-handbook/0.1a/index.html>

3. Install gcc compiler using Alpine commands to install C language in your system.
4. Use wget command to download sample data 1 and sample data 2 in your Alpine VM. these commands are:

Command 1 (Help to download Sample Data1):

wget <https://raw.githubusercontent.com/siddharths-del/TVP-Assignments/main/MCASampleData1.tar.gz>

Command 2 (Help to download Sample Data2):

wget <https://raw.githubusercontent.com/siddharths-del/TVP-Assignments/main/MCASampleData2.tar.gz>

5. After downloading please extract the compressed files to use the data.

Problem Statement:

Question 1:

- a. Analyze the data stored in the files present in the extracted folders namely “MCASampleData1” and “MCASampleData2”. Explain your analysis.

[Note: These are only 2 sample folders. You may have more data in the same format with different headers.

For example

Name: Radha,

Roll:26,

In this {Name and Roll} are headers and {Radha and 26} are data or values allocated to those headers.

]

- b. Based on your analysis (i.e. in question 1 (a)) design a data structure that can store data present in files and then design a system to perform following operations using your data structure:
 - i. Sorting w.r.t specific header.
 - ii. Insert new record.
 - iii. Delete existing record.
 - iv. Update existing record.

[Note 1: All operations (i) to (iv) must be performed in RAM and not in files. However, results must be reflected back in files based on user demand. It means once data is loaded in your data structure you can iteratively perform (i) to (iv) operations without affecting data present in files. Once the above operations are performed and the user wants to store the result on persistent storage then only changes are stored in files.

Note 2: The designed system must be generic. It means if we create a new data pool (say MCASampleData3) with different headers (say {Name,age,fathername,mothername,course,etc.}) then your system must perform the above mentioned operations on the newly created data pool.

Hint: Use concepts like templates, shell script or gcc (to recompile the part of code) and processes in C (to run shell script or direct gcc).

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Question 2:

- a. Update the system designed in Question 1 to perform Join operations (like full join, inner join, outer join, etc.).
- b. Design your own semi-structured query language using functions designed in Question 1 and Question 2 (a). This query language must perform some basic selection, projection and join based operations. [Hint: (i) you are free to choose the query language syntax, (ii) all computations must be done on RAM using your data structure, (iii) after computation the result must be stored in the form of files].
- c. Calculate time required to execute operations performed in Question 1 (b) and to run queries designed in question 2 (a) and (b). [Hint: only execution time is to be calculated. Please exclude the time taken to get input from the user. Use the C library to calculate execution time.]

Note: Explain your answers clearly using text, figures, graphs and other presentation materials. Explain your codes, system design and system architecture.

My dear students, please try to do this assignment in teams without using AI tools to enhance your coding and cognitive skills. I hope you all learn something new from this assignment. All the very best to all. Please feel free to call me at 9455280244 for any queries.

Regards,
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